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Assignment on

Complete Implementation & Analysis of the Playfair Cipher

Course: Cybersecurity Fundamentals
Department of Computer Science and Engineering
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Exercise 2.1 — Matrix Construction

In a Playfair cipher, the 5×5 key matrix is built by taking a keyword, converting any 'J' to 'I', removing duplicate letters, and filling the remaining slots with the remaining letters of the alphabet (excluding 'J').

For the keyword 'CRYPTOGRAPHY':

1. Convert J to I (no J present): 'CRYPTOGRAPHY.'

2. Remove duplicates: C, R, Y, P, T, O, G, A, H (Note: 'R', 'P', 'Y' are discarded as duplicates)

3. Fill the rest of the alphabet (excluding 'J'): B, D, E, F, I, K, L, M, N, Q, S, U, V, W, X, Z

This yields the following 5×5 key matrix, verified using the build_matrix() function:

	1	2	3	4	5
1	C	R	Y	P	T
2	O	G	A	H	B
3	D	E	F	I	K
4	L	M	N	Q	S
5	U	V	W	X	Z

Exercise 2.2 — Encryption Verification

We verify the step-by-step encryption of the plaintext 'HIDE THE GOLD IN THE TREE STUMP' using the keyword 'PLAYFAIR EXAMPLE'.

Key Matrix for 'PLAYFAIR EXAMPLE' (Duplicates removed, J replaced with I):

	1	2	3	4	5
1	P	L	A	Y	F
2	I	R	E	X	M
3	B	C	D	G	H
4	K	N	O	Q	S
5	T	U	V	W	Z

Detailed Diagram Encryption Steps:

Digram	Matrix Positions	Playfair Rule	Encrypted
HI	H:(2,4), I:(1,0)	Rectangle (Swap Columns)	**BM**
DE	D:(2,2), E:(1,2)	Same Column (Shift Down)	**OD**
TH	T:(4,0), H:(2,4)	Rectangle (Swap Columns)	**ZB**
EG	E:(1,2), G:(2,3)	Rectangle (Swap Columns)	**XD**
OL	O:(3,2), L:(0,1)	Rectangle (Swap Columns)	**NA**
DI	D:(2,2), I:(1,0)	Rectangle (Swap Columns)	**BE**
NT	N:(3,1), T:(4,0)	Rectangle (Swap Columns)	**KU**
HE	H:(2,4), E:(1,2)	Rectangle (Swap Columns)	**DM**
TR	T:(4,0), R:(1,1)	Rectangle (Swap Columns)	**UI**
EX	E:(1,2), X:(1,3)	Same Row (Shift Right)	**XM**
ES	E:(1,2), S:(3,4)	Rectangle (Swap Columns)	**MO**
TU	T:(4,0), U:(4,1)	Same Row (Shift Right)	**UV**
MP	M:(1,4), P:(0,0)	Rectangle (Swap Columns)	**IF**

Final Ciphertext (spaces removed):

BMODZBXDNABEKUDMUIXMMOUVIF

Exercise 2.3 — Diagram and Statistical Analysis

a) Most Common English Digram in the Corpus

The most common English digraph in our corpus is 'TH' with a relative frequency of 4.24%. This closely aligns with standard English language statistical models, in which the digram 'TH' represents the most predominant and recurring character pair.

b) Digram Frequency in the Ciphertext

No, 'TH' is no longer the most common digram in the ciphertext. In fact, the most common ciphertext digram is now 'PD' with a frequency of only 2.92%. The sharp peaks of the original language distribution are heavily flattened, and the original letter-pair statistical footprints are thoroughly diffused across the alphabet. This highly uniform and scattered distribution demonstrates the robust polyalphabetic substitution strength of the Playfair cipher.

c) Index of Coincidence (IC) Evaluation

We mathematically calculated the Index of Coincidence (IC) for both the plaintext and ciphertext:

Plaintext Index of Coincidence (IC):	**0.06384**
Ciphertext Index of Coincidence (IC):	**0.04781**
Theoretical Random Text IC (1/26):	**0.03846**

Observation and Theoretical Rationale:

The plaintext IC (~0.0638) is extremely close to the expected value for standard English text (~0.0667), which reflects a highly structured, non-random distribution of natural language letters. In contrast, the ciphertext IC (~0.0478) is substantially lower, shifting significantly closer to the theoretical value of a perfectly random text (1/26 = 0.0385).

By encrypting pairs of letters (digrams) rather than individual characters, the Playfair cipher effectively neutralizes simple monoalphabetic frequency analysis. The substantial reduction in the Index of Coincidence proves mathematically that the ciphertext presents a high degree of entropy (randomness), verifying the cipher's superior cryptographic resistance relative to simpler substitution methods.

PT-109 Decryption Walkthrough and Historical Context

During World War II, Lieutenant (and future US President) John F. Kennedy's boat, PT-109, was rammed and sunk by a Japanese destroyer in the Solomon Islands. A famous rescue mission ensued. A Playfair cipher message was transmitted by the Royal New Zealand Navy to coordinate search and rescue efforts.

We verified the decryption of this historical ciphertext using our implementation:

Key: 'ROYAL NEW ZEALAND NAVY'
Ciphertext: KXJEY UREBE ZWEHE WRYTU HEYFS KREHE GOYFI WTTTU OLKSY CAJPO BOTEI ZONTX BYBNT GONEY CUZWR GDSON SXBOU YWRHE BAAHY USEDQ

Decryption Matrix (ROYAL NEW ZEALAND NAVY):

	1	2	3	4	5
1	R	O	Y	A	L
2	N	E	W	Z	D
3	V	B	C	F	G
4	H	I	K	M	P
5	Q	S	T	U	X

Decrypted Plaintext:

PTBOATONEOWENINELOSTINACTIONINBLACKESSSTRAITTWOMILESSWM ERESUCOVEXCREWOFTWELVEXREQUESTANYINFORMATIONX

Analysis of Field-Encryption Anomalies:

Historically, the operator made spelling and encryption errors while encoding the message in the field. Our programmatic decryption successfully reconstructs these exact anomalies:

1. 'OWENINE' instead of 'ONE O NINE':

The word 'ONE' was incorrectly encrypted/decrypted, yielding 'OWENINE,' which telegraphically reads as 'PT BOAT ONE O NINE'.

2. 'BLACKESSSTRAIT' instead of 'BLACKETT STRAIT':

Because the double-T in 'BLACKETT' required duplicate padding during encryption, and due to operational slip-ups in the field, the letters decrypted to 'BLACKESSSTRAIT' rather than 'BLACKETTSTRAIT'.

Despite these manual errors, the message remained highly comprehensible and was successfully deciphered by search operators, leading directly to the safe rescue of the PT-109 crew. This real-world event perfectly illustrates the robustness of the Playfair cipher against single-character corruptions and its extreme resilience under imperfect field conditions.

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